

QUESTION DESCRIPTION

Your HR chatbot retrieves relevant policy paragraphs but still returns raw text to employees. Wire up the generation step so it produces a plain-English answer from the retrieved passages.

Task

In this task, you will implement the answer generation stage of a RAG pipeline: formatting a prompt from a query and retrieved context, then calling an LLM to produce a grounded response.

- Prompt Building
 - Format the user query and retrieved context passages into a single prompt string.
 - The prompt must instruct the LLM to answer using only the provided context and to be concise.
- Answer Generation
 - Call the LLM with the formatted prompt and return the generated answer.

Note: You are not expected to retrieve documents or build a vector store; only prompt formatting and the LLM call are required.

Deliverables

File	Description
answer_generator.py	prompt formatting and LLM-based answer generation

Sample Policy Passages

All full-time employees are entitled to 20 days of paid annual leave per calendar year.
Business expenses up to £50 may be submitted without prior approval.
Performance reviews are conducted twice a year, in June and December.
Primary caregivers are entitled to 26 weeks of paid parental leave at full salary.
Employees must complete mandatory health and safety induction training within their first week.


```

1 from litellm import completion
2 from typing import List
3
4
5 class AnswerGenerator:
6     """Generates a natural-language answer using retrieved context and an LLM via litellm."""
7
8     def __init__(self, model: str = "claude-haiku-4-5-20251001") -> None:
9         """Store the litellm model identifier as an instance attribute."""
10         self.model = model
11
12     def build_prompt(self, query: str, context: str) -> str:
13         """Format query and context into a prompt string; raise ValueError if query is empty."""
14         if not query or not query.strip():
15             raise ValueError("Query cannot be empty.")
16
17     def generate(self, query: str, context: str) -> str:
18         """Build the prompt, call litellm.completion, and return the answer string; raise ValueError
19         if query is empty."""
20         raise NotImplementedError("Implement this method")
21

```


Question 1

You are analyzing the performance of a dual cache system that processes n service requests.

For each request i :

- $\text{payloadSize}[i]$ is the size of the request.
- $\text{cacheA}[i] = 1$ if cache A can serve request i , otherwise 0.
- $\text{cacheB}[i] = 1$ if cache B can serve request i , otherwise 0.

You want to select a subset of requests such that:

- Cache A can serve at least minThreshold of the selected requests.
- Cache B can serve at least minThreshold of the selected requests.

Among all such valid subsets, you must find the one with the minimum possible sum of payload sizes.

Return:

- the minimum sum of payload sizes of such a subset, or
- -1 if no such subset exists.

Example 1

Suppose $n = 6$, $\text{payloadSize} = [10, 8, 12, 4, 5, 25]$, $\text{cacheA} = [1, 0, 1, 1, 0, 1]$, $\text{cacheB} = [1, 0, 1, 0, 1, 1]$, and $\text{minThreshold} = 3$.

Suppose $n = 6$, $payloadSize = [10, 8, 12, 4, 5, 23]$, $cacheA = [1, 0, 1, 1, 0, 1]$, $cacheB = [1, 0, 1, 0, 1, 1]$, and $minThreshold = 3$.
Output: 31

If we choose the subset [1st, 3rd, 4th, 5th] requests:

- Cache A can serve the 1st, 3rd, and 4th requests (3 requests, meeting the threshold)
- Cache B can serve the 1st, 3rd, and 5th requests (3 requests, meeting the threshold)
- Sum of payload sizes: $10 + 12 + 4 + 5 = 31$

This is the minimum possible sum that satisfies the constraints.

Example 2

Suppose $n = 5$, $payloadSize = [3, 2, 4, 1, 5]$, $cacheA = [0, 0, 0, 0, 1]$, $cacheB = [1, 1, 0, 1, 1]$, and $minThreshold = 2$.
Output: -1

Cache A can only serve one of the five requests, which is lower than the $minThreshold$ of two. Therefore, it is not possible to serve the requests from both caches.

Constraints

- $1 \leq n \leq 2 * 10^5$
- $1 \leq payloadSize[i] \leq 10^9$
- $0 \leq cacheA[i], cacheB[i] \leq 1$
- $1 \leq minThreshold \leq n$.




```

14 > import java.io
15 class Result {
16
17     /*
18      * Complete the 'getMinimumSize' function below.
19      *
20      * The function is expected to return a LONG INTEGER.
21      * The function accepts following parameters:
22      * 1. INTEGER_ARRAY payloadSize
23      * 2. INTEGER_ARRAY cacheA
24      * 3. INTEGER_ARRAY cacheB
25      * 4. INTEGER minThreshold
26      */
27
28     public static long getMinimumSize(List<Integer> payloadSize, List<Integer> cacheA,
29     List<Integer> cacheB, int minThreshold) {
30         // Write your code here
31     }
32
33 }
34
35 > public class Solution { ...

```


Question 3

Customers email the bank to report transaction disputes. Before creating a dispute ticket, the system must verify that the email contains all mandatory details.

The following information must be present:

- Customer name
- Transaction amount
- Transaction date
- Last four digits of card

Task

Write prompts to query customer emails and determine whether required information is missing.

If all fields are present:

COMPLETE

If any are missing:

MISSING: field1,field2

Allowed field names:

- name
- amount
- date
- card_last_four

Language: Markdown

- 1 Write your prompt here.
- 2
- 3 {testcase input}

^ Test Results

46 min

- card_last_four

Rules:

- If multiple fields are missing, list them in alphabetical order.
- Separate fields using comma (no spaces).
- Output must exactly match the required format.
- Do not include explanations.

NOTE: The {testcase input} field is a placeholder that will be auto-filled with various inputs to test the prompt.

▼ Sample Case 1

Sample Input

Subject: Unrecognized Charge on My Statement

Hello,

While reviewing my September statement this morning, I noticed a charge of \$189.60 dated September 14, 2024 that I don't recall making. The last four digits of my card are 8821.

Could you please look into this?

Kind regards,
Lauren Mitchell

Sample Output

COMPLETE



1 hr 30 min



COMPLETE

Language

Explanation

All required fields present:

- name → Lauren Mitchell
- amount → 189.60
- date → September 14, 2024
- card_last_four → 8821

▼ Sample Case 2

Sample Input

Subject: Something Looks Wrong

Hi,

There is a transaction I don't recognize.
My card ends in 2299.

Thanks,
Noah Bennett

Sample Output

MISSING: amount, date

Explanation

Amount and date patterns are missing.

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